

Predicting Neighborhood Socioeconomic Upgrading: ML for Gentrification Early Warning

Research Proposal · May 2025

1. Problem

Gentrification — socioeconomic upgrading of previously low-income neighborhoods — is identified only *after* Census data records the change. Policymakers need early warning. Can ML predict which census tracts will undergo upgrading *before* it happens?

Two methodological problems undermine existing ML gentrification prediction:

1. **Circularity.** Prior studies construct the gentrification label from ACS variables (income, rent, home value, education), then use the *same variables* at T1 as input features. Reported accuracy conflates genuine prediction with regression-to-the-mean detection.
2. **Definition dependence.** No consensus definition exists. “Variable selection had a significant impact on which, and how many, census tract areas were defined as neighbourhoods undergoing gentrification” (Galster & Peacock 1986). Studies report one accuracy number under one definition, obscuring how much is an artifact of label construction.

2. Research Questions

RQ1 — Circularity. How much does reported accuracy come from feature-label overlap? What is the AUC gap between models with and without label-defining variables as features?

RQ2 — Leading indicators. Is T1 home value a genuine leading indicator of income-based gentrification (per Buntén et al. 2024’s expectations theory), or a mechanical correlate? The feature-label separation design creates a clean test: home value is excluded under composite definitions (where it defines the label) but allowed under income-only definitions (where it does not).

RQ3 — Definition sensitivity. How do model performance and tract classification vary across five operationalizations? Where do definitions agree (consensus tracts) and disagree (contested tracts)?

RQ4 — Spatial autocorrelation. How much does ignoring spatial dependence inflate CV estimates? Does a spatial-proximity heuristic (“tracts adjacent to already-gentrified tracts will gentrify”) outperform ML?

3. Approach

3.1 Study Design

- **City:** Chicago (800 census tracts). Single city chosen for data quality and statistical power.
- **Temporal structure:** T1 = ACS 2010–2014, T2 = ACS 2015–2019 (non-overlapping, both on 2010-vintage tract boundaries).
- **Task:** Binary classification — predict T2 gentrification label from T1 features.
- **Expected prevalence:** 10–15% (80–120 positive cases).

3.2 Five Gentrification Definitions

All share an eligibility criterion (T1 income < metro median) but differ in what constitutes “change.” **D4 (Composite) is the primary specification; others are sensitivity analyses.**

ID	Name	Change criteria	Source
D1	Freeman	Education increase > metro	Freeman (2005)
D2	Ding-Hwang-Divringi	Rent/ value + education > metro	Ding et al. (2016)
D3	Income-only	Income growth > metro	Simplified baseline
D4	Composite	Income + rent/ value + education > metro	Adapted from NCRC
D5	Expectations-based	House-value pctlte surpasses income pctlte	Buntén et al. (2024)

3.3 Primary Contribution: Definition-Specific Feature-Label Separation

Rather than blanket-excluding all ACS variables, we exclude a variable from the feature set *if and only if it appears in that definition's label*. This creates a natural experiment:

- Under **D3** (income-only): home value, rent, and education are *allowed* as features.
- Under **D4** (composite): all three are *excluded*.

Comparing D3 vs. D4 separates the **definition effect** (different tasks) from the **feature-availability effect** (more / fewer features). If home value ranks high in SHAP importance under D3, it is a genuine leading indicator — not circularity — because it does not define D3's label.

3.4 Feature Set

25 always-included features spanning demand-side signals (demographics, transit, schools, amenities) and supply-side signals (vacancy, housing age, permits, zoning). Up to 3 conditionally-included features depending on definition. All from public sources: ACS, Chicago open data, CTA GTFS, NCES, HUD. See Appendix C.

3.5 Models & Evaluation

Model	Purpose
Spatial proximity heuristic	Baseline: does ML beat spatial diffusion?
L1 logistic regression	Interpretable linear baseline
Random Forest	Non-linear, standard in literature
XGBoost	Typically strongest on tabular data

Evaluation: 5-fold stratified CV and 5-fold spatial block CV reported side-by-side. Probability calibration (Platt scaling) on unweighted models. All metrics with 95% bootstrap CIs.

3.6 Four Experiments

#	Question	Design
1	Circularity quantification	With/ without label-defining features; report AUC gap
2	Leading indicator test	Compare SHAP rankings for home value under D3 vs. D4
3	Racial composition ablation	With/ without racial features; test proxy effects
4	Supply vs. demand ablation	Supply-only, demand-only, and combined feature sets

4. Contributions

1. **Definition-specific feature-label separation** — no prior ML gentrification study implements this. Creates a clean test of circularity vs. genuine prediction.
2. **Definition sensitivity analysis** — systematic comparison across five operationalizations, extending the NCRC (2025) definition comparison from descriptive to predictive.
3. **Spatial CV + calibration** — exposes autocorrelation inflation and produces calibrated risk probabilities for policy use.

Either outcome is publishable: accuracy holds under separation → genuine early-warning signals exist; accuracy drops → prior work was inflated by circularity.

5. Timeline (10 Weeks)

Week	Focus	Deliverable
1–2	Definitions, literature, data audit, labels, agreement	Label dataset, agreement matrix, EPV report
3–4	Data pipelines (ACS, city data, transit, schools) & features	Merged analysis-ready dataset
5–6	Baselines, RF, XGBoost, spatial block CV, calibration	Primary results (D4)
7–8	Four experiments, SHAP, sensitivity (D1–D3, D5), failure analysis	Circularity & sensitivity tables
9–10	Maps, visualization, paper writing	Draft manuscript

6. Key Risks

Risk	Why it matters
Low EPV (3–5)	80–120 positives with 25 features. Mitigated by a priori feature count, stability analysis, Riley et al. sample size check.
Single temporal transition	No out-of-time validation. Great Recession recovery may masquerade as gentrification. Highest-priority future extension.
Upgrading $\neq$ displacement	Tract-level averages shift without any individual being displaced (ecological fallacy). Model cannot distinguish in-situ upgrading from population replacement.
Dual-use	Risk maps could be used by speculators. Framed around community protection; ethics discussion included.

7. Fit

- Builds directly on Anna’s existing census-tract research in Chicago (Cornell, 2024).
- All data publicly available — no API dependencies, no scraping.
- Targets *Environment and Planning B*, *CEUS*, *Housing Policy Debate*, *Urban Studies*.

Appendices

A: Gentrification Definitions

The Measurement Problem

“No agreed-upon definition has become paradigmatic” (Desmond, An & Xiang 2023). The binary threshold approach is “criticized for the arbitrariness of the thresholds” (Li et al. 2019). Rather than evaluating definitions against ground truth (which does not exist), we test how model performance and tract classification vary across definitions.

ID	Eligibility	Change Criteria	Label-Defining Vars
D1	Central city, income < median, older housing	Education increase > metro	Income, education
D2	Income < median	Rent OR value > metro, AND education > metro	Income, rent, value, education
D3	Income < median	Income growth > metro	Income only
D4	Income < median	Income + rent/value + education > metro	All four
D5	Income < median	Value pctlte surpasses income pctlte by ≥ 25 pp	Income, home value

Agreement analysis: For each pair, we report Cohen’s κ , PABAK (Byrt et al. 1993) with 95% bootstrap CIs, and consensus vs. contested tracts. AUC under different definitions measures different tasks — we compare *within*-definition only.

The Expectations Channel (D5)

Bunten et al. (2024): “Property values today incorporate market participants’ expectations of the neighbourhood’s future.” When a tract’s house-value percentile rises above its income percentile, the *market* expects gentrification before demographics register it. This gap “predicts future income growth” with a 3-year lead. T1 home value is therefore a **theoretically justified leading indicator** — whether it belongs in the feature set depends on whether it appears in the label.

B: Feature-Label Separation Design

Variable Taxonomy

1. Definitional proxies — algebraic transforms of label-defining variables. Excluded under ALL definitions:

- Poverty rate (monotonic transform of income), % receiving public assistance (income-tested eligibility).

2. Label-defining variables — excluded only under definitions that use them:

Variable	Excluded under	Allowed under
Median household income	All (eligibility)	None
Median gross rent	D2, D4	D1, D3, D5

Variable	Excluded under	Allowed under
Median home value	D2, D4, D5	D1, D3
% bachelor's degree+	D1, D2, D4	D3, D5

3. Economic correlates — correlated with income but structurally independent (% renter-occupied, vacancy rate, unemployment rate). Included with justification.

The Natural Experiment

Under D3 the feature set grows to 28 variables (25 + rent + home value + education); under D4 it stays at 25. If D3 performs substantially better *and* home value ranks high in SHAP importance under D3, home value is a genuine leading indicator (per Bunten et al.) — not circularity.

Mutual information audit: Before modeling, compute MI between each candidate feature at T1 and (a) each label-defining variable at T1, and (b) the binary label. Permutation-based threshold ($\alpha=0.05$). Flagged features investigated, not automatically excluded.

C: Feature Set

Always-Included (25 variables)

Category	Features	N	Role
Demographics	% White, Black, Hispanic, foreign-born, median age	5	Demand
Housing (non-price)	% renter, % vacant, housing age (% pre-1960), % moved in 5 yrs	4	Supply
Employment	% service occupations, mean commute time	2	Demand
Subsidized housing	HUD Housing Choice Voucher count	1	Policy
Zoning	Dominant class, max density	2	Supply
Urban signals	Crime rate, building permits, 311 complaints, business licenses	4	Mixed
Transit	Distance to nearest CTA L station, L stations within 800m	2	Demand
School quality	Mean K–12 test score percentile	1	Demand
Amenity	Distance to park, distance to university	2	Demand
Spatial	Distance to CBD, population density	2	Context

Features grounded in two explanatory camps: **demand-side** (Ley 1986: preferences of a new middle class) and **supply-side** (Smith 1979: rent gap theory — where the gap between current and potential land rent is largest).

Conditionally-Included

Median gross rent (allowed under D1, D3, D5), median home value (allowed under D1, D3), % bachelor's degree+ (allowed under D3, D5).

Data Sources Excluded

Zillow ZHVI (ZIP-level only), Yelp/Google Places (no historical data → temporal leakage), Google Street View / satellite (infeasible in 10 weeks), LEHD LODES (complex pipeline, EPV constraints). All flagged as future extensions.

D: Sample Size & Statistical Power

EPV: 80–120 positives with 25 features → EPV $\approx$ 3.2–4.8. Under D1 (restrictive eligibility), positives may drop to 40–60 (EPV $\approx$ 1.6–2.4). The classic EPV ≥ 10 rule rests on weak evidence (van Smeden et al. 2016). We compute Riley et al. (2019) minimum sample size at the outset.

Implications: Feature importance rankings will be unstable; tree defaults will overfit; L1 reduces effective count but not search space.

Mitigations: (1) A priori feature count of 25. (2) Riley et al. computation reported explicitly. (3) Conservative hyperparameters (high `min_samples_leaf`, low `max_depth`); no nested CV. (4) Stability analysis: 50 seeds, report Spearman ρ of SHAP rankings (if $\rho < 0.7$, state explicitly). (5) Bootstrap CIs ($n=1000$); if AUC CI > 0.15 , note insufficient sample. (6) Definitions yielding < 30 positives → descriptive only.

Why Chicago only: Baltimore (200 tracts, 20–30 positives) would yield 4–6 positives per fold. Chicago’s open data is substantially more complete.

E: ML Methods & Experiments

Cross-validation: Stratified 5-fold and spatial block CV (block size exceeds Moran’s I autocorrelation range) reported side-by-side. Literature shows spatial CV inflation up to 28% (Karasiak et al. 2022).

Class imbalance: No SMOTE. Two model versions: (a) unweighted (for calibration/risk maps), (b) `class_weight='balanced'` (for discrimination metrics).

Calibration: Platt scaling on unweighted model only. Per Phelps et al. (2024), Platt scaling after reweighting produces biased calibration.

Metrics: AUC-ROC, AUC-PR, precision, recall, F1 (weighted model). Brier score, reliability diagram (unweighted + calibrated). All with 95% bootstrap CIs.

SHAP analysis: Pairwise Pearson $|r| > 0.7 \rightarrow$ cluster features. Report SHAP per-feature and per-cluster (summing $|SHAP|$). Expected clusters: {renter, housing age, vacancy}; {foreign-born, Hispanic}; {crime, 311}; {transit, CBD distance}. SHAP reported only for definitions with EPV ≥ 3 .

Failure analysis: Mispredicted tracts examined for clustering in high-foreclosure areas (Great Recession confound).

F: Temporal & Data Considerations

Why 2010–2014 $\rightarrow$ 2015–2019: Non-overlapping 5-year periods on 2010-vintage boundaries, avoiding NHGIS harmonization. Census Bureau “strongly recommends against comparing estimates in overlapping 5-year periods.”

Great Recession confound: T1 captures post-recession aftermath. Income growth to T2 partly measures *recovery*, not gentrification. Compounded by historically low interest rates (near zero through 2015). Mitigations: acknowledge explicitly; examine misprediction clustering; note all definitions compare to metro-level growth; flag second temporal period as future extension.

ACS margins of error: MOE columns downloaded; tracts where MOE $> 30\%$ of estimate for label-defining variables are flagged.

G: Ethical Considerations

Dual-use: Risk maps could be used by speculators to front-run neighborhood change (Weber 2019: “predictions become self-fulfilling prophecies”). Frame around community protection; avoid publishing interactive tract-level maps without access controls.

Prediction $\neq$ prescription: SHAP shows feature *importance*, not causal mechanisms. “Building permits predict gentrification” $\neq$ permits *cause* gentrification.

Racial features: If racial composition is a top predictor, it may proxy for structural disadvantage (redlining, disinvestment). Experiment 3 tests this directly.

Ecological fallacy: A “gentrifying” tract may reflect new construction, in-situ income growth, or population replacement. We predict *socioeconomic upgrading*, not *displacement*. The relationship between the two is empirically contested (Freeman 2005; Desmond et al. 2023; Brummet & Reed 2019 vs. Newman & Wyly 2006; Betancur 2011).

H: Prior Work & Positioning

Study	Method	Result	How we differ
Census Bureau WP 2023	RF, D.C.	83% accuracy	No feature-label separation
SSRN 3911354	ACS+Zillow+HUD	74% accuracy	ZIP-level, single def., no spatial CV
Yao et al. (2023)	GNN+satellite	0.9 AUC	More complex; we add cleaner separation

Study	Method	Result	How we differ
Thackway et al. (2023)	RF+SHAP+CNN	74.7% bal. acc.	Single definition, no spatial CV
Bunten et al. (2024)	Expectations	Value/income gap predicts growth	We incorporate as D5
NCRC & UMich (2025)	ROC of defs.	“Depends on measurement”	We test how defs affect <i>prediction</i>

I: Team & Publication

Team: Student A (Census API, ACS MOE, MI audit, transit/amenity/school pipeline), Student B (city open data — crime, permits, 311, licenses, zoning), Student C (ML modeling, spatial CV, SHAP, calibration, maps), Anna (research design, definition comparison, literature, agreement analysis, paper writing, ethics).

Targets: *Environment and Planning B* ★★★★★, *CEUS* ★★★★★, *Housing Policy Debate* ★★★★, *Urban Studies* ★★★★★.

J: References

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